

Problem Set 6

Kerri Riley

```
# Import modules  
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'tibble' was built under R version 4.5.2

Warning: package 'tidyr' was built under R version 4.5.2

Warning: package 'readr' was built under R version 4.5.2

Warning: package 'purrr' was built under R version 4.5.2

Warning: package 'dplyr' was built under R version 4.5.2

Warning: package 'lubridate' was built under R version 4.5.2

-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --

v dplyr 1.2.1 v readr 2.2.0

v forcats 1.0.1 v stringr 1.6.0

v ggplot2 4.0.3 v tibble 3.3.1

v lubridate 1.9.5 v tidyr 1.3.2

v purrr 1.2.2

-- Conflicts ----- tidyverse_conflicts() --

x dplyr::filter() masks stats::filter()

x dplyr::lag() masks stats::lag()

i Use the conflicted package (<<http://conflicted.r-lib.org/>>) to force all conflicts to become

```
library(haven)

# Read the GSS data
gss <- read_dta("~/Documents/PAI741/Problem Set 6/Data/gss7224_r1.dta")
```

Recoding for Replication

```
# spkath, colath, libath
gss <- gss|>
  mutate(sp_kath = case_when(
    sp_kath == 1 ~ 0,
    sp_kath == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(colath = case_when(
    colath == 4 ~ 0,
    colath == 5 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(libath = case_when(
    libath == 1 ~ 0,
    libath == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

# Racists
# spkrac, colrac, librac
gss <- gss|>
  mutate(sp_krac = case_when(
    sp_krac == 1 ~ 0,
    sp_krac == 2 ~ 1,
    # The rest is missing
```

```

    TRUE ~ NA
  ))

gss <- gss|>
  mutate(colrac = case_when(
    colrac == 4 ~ 0,
    colrac == 5 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(librac = case_when(
    librac == 1 ~ 0,
    librac == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

# Communists
# spkcom, colcom, libcom
gss <- gss|>
  mutate(spkcom = case_when(
    spkcom == 1 ~ 0,
    spkcom == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(colcom = case_when(
    colcom == 4 ~ 0,
    colcom == 5 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(libcom = case_when(
    libcom == 1 ~ 0,
    libcom == 2 ~ 1,

```

```

    # The rest is missing
    TRUE ~ NA
  ))

# militarists
# spkmil, colmil, libmil
gss <- gss|>
  mutate(spkmil = case_when(
    spkmil == 1 ~ 0,
    spkmil == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(colmil = case_when(
    colmil == 4 ~ 0,
    colmil == 5 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(libmil = case_when(
    libmil == 1 ~ 0,
    libmil == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

# Homosexuals
# spkhomo, colhomo, libhomo
gss <- gss|>
  mutate(spkhomo = case_when(
    spkhomo == 1 ~ 0,
    spkhomo == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

```

```

gss <- gss|>
  mutate(colhomo = case_when(
    colhomo == 4 ~ 0,
    colhomo == 5 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(libhomo = case_when(
    libhomo == 1 ~ 0,
    libhomo == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

```

Replicate Figure 1

```

# Replicate figures
# Figure 1: 0-15 GSS Modified Stouffer Tolerance Battery, 1976-2024

# Create intolerance score by adding together the sum of all the values for the respondent.
gss <- gss |>
  mutate(intolerance_score = spkath + colath + libath +
    spkrac + colrac + librac +
    spkcom + colcom + libcom +
    spkmil + colmil + libmil +
    spkhomo + colhomo + libhomo)

# Take the mean score of all respondents by year
figure1 <- gss |>
  group_by(year) |>
  summarize(mean_intolerance = mean(intolerance_score, na.rm = TRUE))

# Take figure1 and filter
figure1 <- figure1 |>
  filter(!is.nan(mean_intolerance))

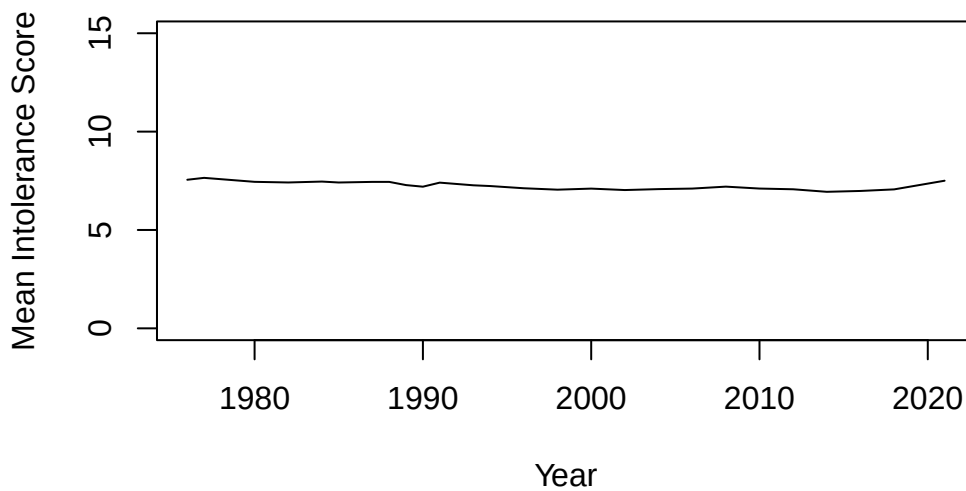
# Show table of data for figure1
figure1

```

```
# A tibble: 26 x 2
  year      mean_intolerance
  <dbl>+<lbl>      <dbl>
1 1976             7.55
2 1977             7.65
3 1980             7.45
4 1982             7.41
5 1984             7.46
6 1985             7.41
7 1987             7.44
8 1988             7.44
9 1989             7.28
10 1990            7.20
# i 16 more rows
```

```
# Create plot 1
plot(figure1$year, figure1$mean_intolerance,
     # Setting theplot type to a line plot
     type = "l",
     # Main title of figure
     main = "The 0-15 GSS Modified Stouffer Tolerance Battery, 1976-2024",
     # X-axis label
     xlab = "Year",
     # Y-axis label
     ylab = "Mean Intolerance Score",
     # Parameters for y-axis
     ylim = c(0,15)
)
```

The 0–15 GSS Modified Stouffer Tolerance Battery, 1976–20



Replicate Figure 2

```
# Figure 2: Intolerance Towards Racists and Homosexuals, 1976–2024

gss <- gss |>
  mutate(
    racist_score = spkrac + colrac + librac,
    homosexual_score = spkhomo + colhomo + libhomo
  )

figure2 <- gss |>
  group_by(year) |>
  summarize(
    racist_mean = mean(racist_score, na.rm = TRUE),
    homosexual_mean = mean(homosexual_score, na.rm = TRUE)
  )

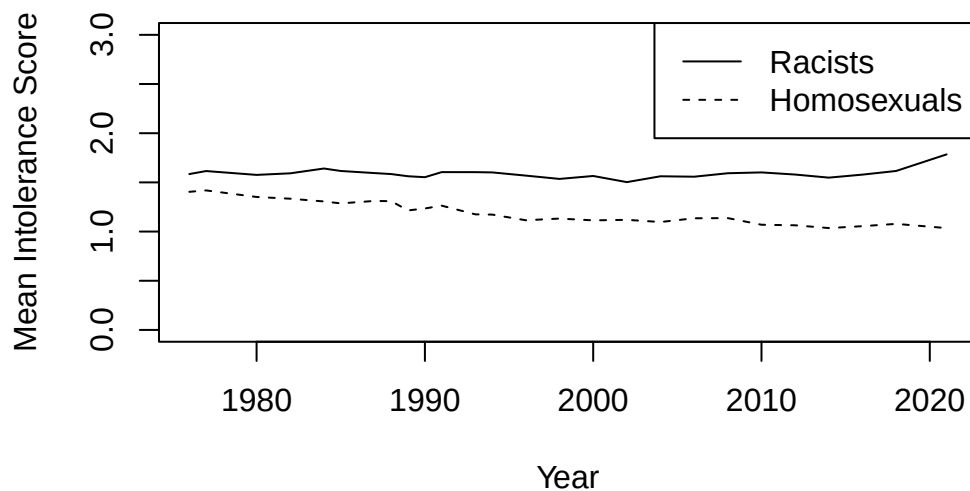
# Remove years without data
figure2 <- figure2 |>
  filter(!is.nan(racist_mean) & !is.nan(homosexual_mean))
```

```
# Create plot
plot.figure2$year, figure2$racist_mean,
      type = "l",
      ylim = c(0, 3),
      main = "Intolerance Towards Racists and Homosexuals, 1976-2024",
      xlab = "Year",
      ylab = "Mean Intolerance Score")

# Add homosexual intolerance line
lines.figure2$year, figure2$homosexual_mean,
      lty = 2)

# Add legend
legend("topright",
      legend = c("Racists", "Homosexuals"),
      lty = c(1, 2))
```

Intolerance Towards Racists and Homosexuals, 1976-2024



Replicate Figure 5

```

# Figure 5: A Dichotomous Measure of Tolerance and Intolerance, 1976-2024
# If a respondent gives at least one intolerant response among the 15 items, they are classi

# Create dichotomous measure

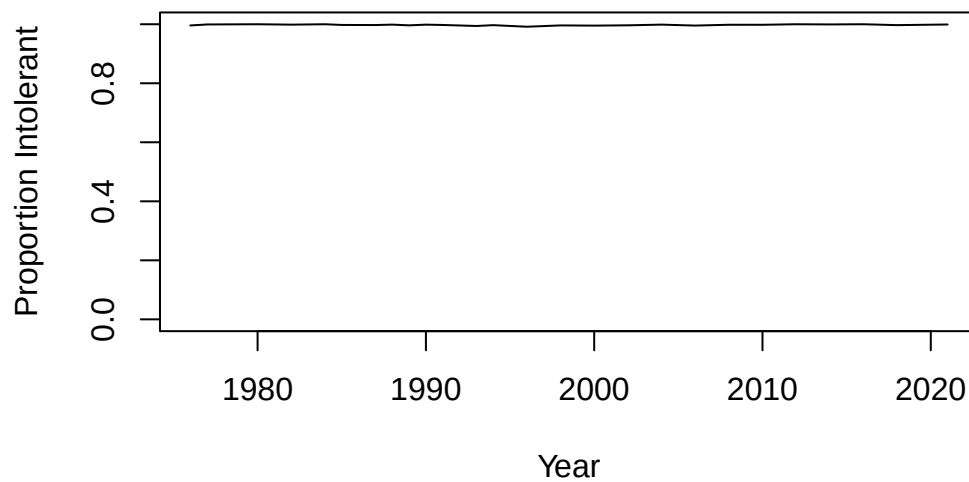
gss <- gss |>
  mutate(intolerant = case_when(
    intolerance_score == 0 ~ 0,
    intolerance_score > 0 ~ 1,
    # If there is a missing value for intolerance_score, keep it missing
    TRUE ~ NA
  ))

# Calculate proportion intolerant by year
figure5 <- gss |>
  group_by(year) |>
  summarize(
    proportion_intolerant = mean(intolerant, na.rm = TRUE)
  ) |> # Remove years with no data
  filter(!is.nan(proportion_intolerant))

# Create plot
plot(figure5$year, figure5$proportion_intolerant,
     type = "l",
     ylim = c(0, 1),
     main = "A Dichotomous Measure of Tolerance and Intolerance, 1976-2024",
     xlab = "Year",
     ylab = "Proportion Intolerant")

```

A Dichotomous Measure of Tolerance and Intolerance, 1976-



Original Work

```
# HEALTH

gss <- gss |>
  mutate(health = case_when(
    health == 1 ~ 4,
    health == 2 ~ 3,
    health == 3 ~ 2,
    health == 4 ~ 1,
    # The rest is missing
    TRUE ~ NA_real_
  ))

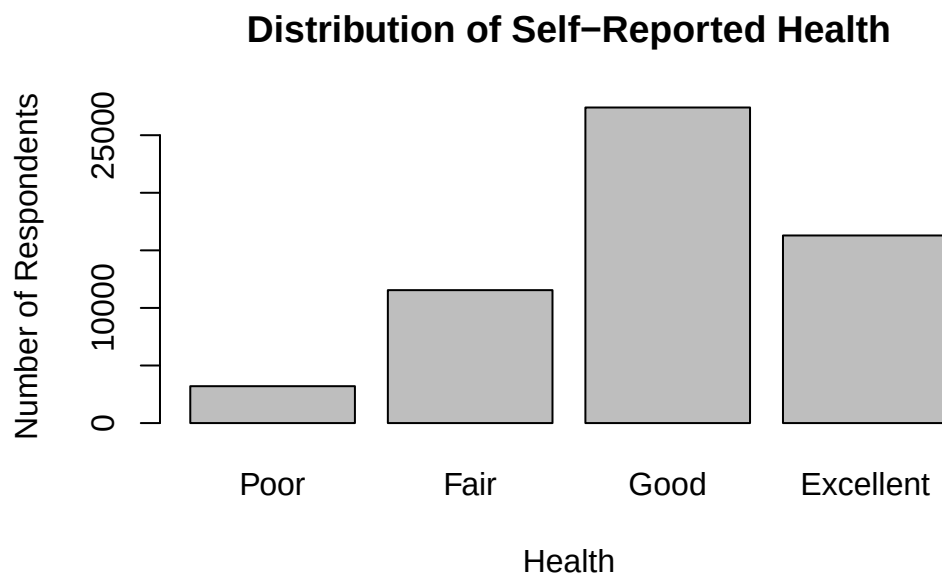
# Central tendency and spread
mean(gss$health, na.rm = TRUE)
```

```
[1] 2.971599
```

```
sd(gss$health, na.rm = TRUE)
```

```
[1] 0.8335249
```

```
# Distribution of Health Responses
barplot(table(gss$health),
        main = "Distribution of Self-Reported Health",
        xlab = "Health",
        ylab = "Number of Respondents",
        names.arg = c("Poor", "Fair", "Good", "Excellent"))
```



```
# HAPPINESS
```

```
gss <- gss |>
  mutate(happy = case_when(
    happy == 1 ~ 3,
    happy == 2 ~ 2,
    happy == 3 ~ 1,
    # The rest is missing
    TRUE ~ NA_real_
  ))
```

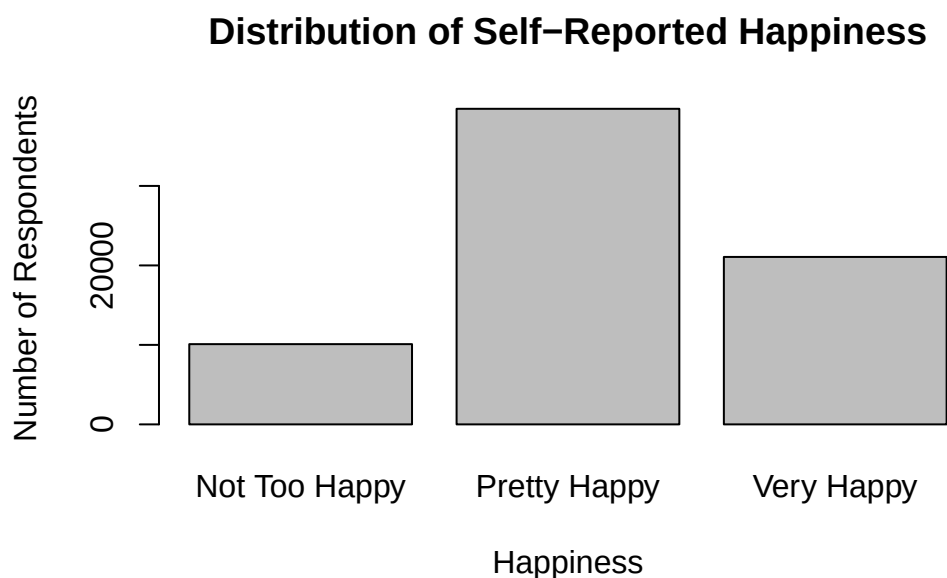
```
# Central tendency and spread  
mean(gss$happy, na.rm = TRUE)
```

```
[1] 2.154849
```

```
sd(gss$happy, na.rm = TRUE)
```

```
[1] 0.6448012
```

```
# Distribution of Happiness Responses  
barplot(table(gss$happy),  
        main = "Distribution of Self-Reported Happiness",  
        xlab = "Happiness",  
        ylab = "Number of Respondents",  
        names.arg = c("Not Too Happy", "Pretty Happy", "Very Happy"))
```



Where did you encounter problems with this replication? Was there anything that was particularly tricky?

I had a difficult time identifying which variables to use in the code book to replicate the figures, as there were over 6,000 columns.

How would you interpret (in other words, explain to a general audience) what Mondak and Sanders found in figures 1, 2, and 5?

Figure 1 Interpretation: According to Mondak and Sanders, the Americans became more tolerant. Figure 1 uses the 0-15 GSS tolerance scale, where a higher score indicates greater intolerance. The average score fell from 7.24 in 1976 to 5.28 in 1998. However, Mondak and Sanders argue that people may simply have become less hostile towards some of the groups included in the survey.

Figure 2: Figure 2 helps explain why the authors are skeptical of Figure 1. They separate responses concerning racists and homosexuals and find very different trends. Intolerance toward homosexuals declined by more than intolerance toward racists.

Figure 5: Figure 5 suggests the change in tolerance is much smaller. Instead of asking how many intolerant responses someone gives, they divide respondents into two categories: someone who gives zero intolerant responses is classified as tolerant, while someone who gives at least one intolerant response is classified as intolerant. The proportion classified as intolerant falls only slightly from about 85% in 1976 and 1988 to 79% in 1998, a decline of less than six percentage points.

Has anything changed since 1998 that would alter the substantive takeaway points in their findings?

My updated figures suggest that Mondak and Sander's main conclusion still holds. My extension shows that from roughly 1998 onward, the overall intolerance score does not continue falling. Instead, it fluctuates around roughly 8 in my replication and even rises somewhat in the most recent observations. So there is little evidence in my extended series of a continuing long-term movement toward greater tolerance after 1998. For figure 2, my extension shows that attitudes towards the two groups continue to move differently rather than converging around a common trend. This reinforces the idea that a singular tolerance score can mask substantial differences in attitudes towards a specific group. Lastly, figure 5 (continues) shows that the proportion remains pretty high and flat, suggesting that the targets of intolerance may have changes, but the proportion of intolerant respondents likely did not change substantially.